Technology Stack

Date	01 October 2026
Team ID	SWTID-2026-1940
Project Name	Team Enrollment Confirmation - EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that EduGenie is scalable, maintainable, secure, and aligned with the project requirements.

The table below details the technologies selected for each architecture component and provides a brief justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** React, HTML/CSS/JavaScript	**React.js** HTML5 CSS3 JavaScript	React provides reusable components and a responsive interface for student enrollment, confirmation status and the EduGenie learning-assistant screens. It supports fast UI development and easy maintenance.
2	**Backend / Server-Side** Node.js / Express	**Node.js** Express.js	Node.js with Express.js provides a lightweight API layer for authentication, enrollment processing, database operations and secure communication with the Gemini API.
3	**Database / Data Storage** MongoDB	**MongoDB**	MongoDB is suitable for storing user profiles, team/student enrollment records, course information, confirmation status and learning-history data using a flexible document structure.

Technology Stack — Continued

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
4	**Cloud / Hosting / Deployment** Web hosting and server deployment	**Vercel** Frontend **Render** Backend	Vercel provides simple deployment and scalable hosting for the React frontend. Render can host the Node.js/Express backend with environment-variable support for secure configuration.
5	**Version Control & CI/CD** GitHub	**Git** GitHub GitHub Actions	Git and GitHub support team collaboration, version tracking, branch-based development and code review. GitHub Actions can automate build and test checks when changes are pushed.
6	**Third-Party APIs / Other Tools** AI learning assistance	**Google Gemini API** Postman	The Google Gemini API powers EduGenie's AI learning assistant by generating educational responses from user prompts. Postman can be used to test and validate backend API endpoints during development.

Technology Selection Summary

Layer	Selected Technology
Frontend	React.js, HTML5, CSS3, JavaScript
Backend	Node.js, Express.js
Database	MongoDB
Deployment	Vercel + Render
Version Control / CI	Git, GitHub, GitHub Actions
AI / API Testing	Google Gemini API, Postman

Team Work Assignment

The responsibilities below are assigned according to the exact order of names provided.

S.No	Team Member	Assigned Work
1	**R.Priya dharshini**	Frontend Lead — React.js UI development, student registration/login screens, enrollment form and responsive page design.
2	**S.Shobana**	Backend Lead — Node.js/Express API development, authentication logic, enrollment processing and backend integration.
3	**S.Rakshana**	Database & Data Management — MongoDB schema/design, user and enrollment records, confirmation status and data validation.
4	**E.Shylaja**	AI Integration & Testing — Google Gemini API integration, prompt handling, learning-assistant features and API testing.
5	**S.Sheeba**	Deployment, Documentation & QA — Git/GitHub workflow, deployment support, project documentation, testing checklist and final review.

Project Members

No.	Name
1	R.Priya dharshini
2	S.Shobana
3	S.Rakshana
4	E.Shylaja
5	S.Sheeba